CHUNKING MEMORY VS RECALL

TITLE:

Design a User Interface where users recall visual elements and evaluate the effect of chunking on memory performance.

AIM:

To design and prototype a memory recall task using chunked visual elements and analyze the effect of chunking on user memory.

SOFTWARE USED:

• Figma (UI Design Tool).

THEORY:

Memory recall refers to the ability of an individual to retrieve previously viewed information from short-term memory. Cognitive psychology suggests that humans can retain a limited number of items in short-term memory.

Chunking is a memory enhancement technique where information is grouped into meaningful units. According to Miller's Law (7 ± 2 rule), individuals can remember approximately 5 to 9 items in short-term memory. Grouping items improves recall efficiency by reducing cognitive load.

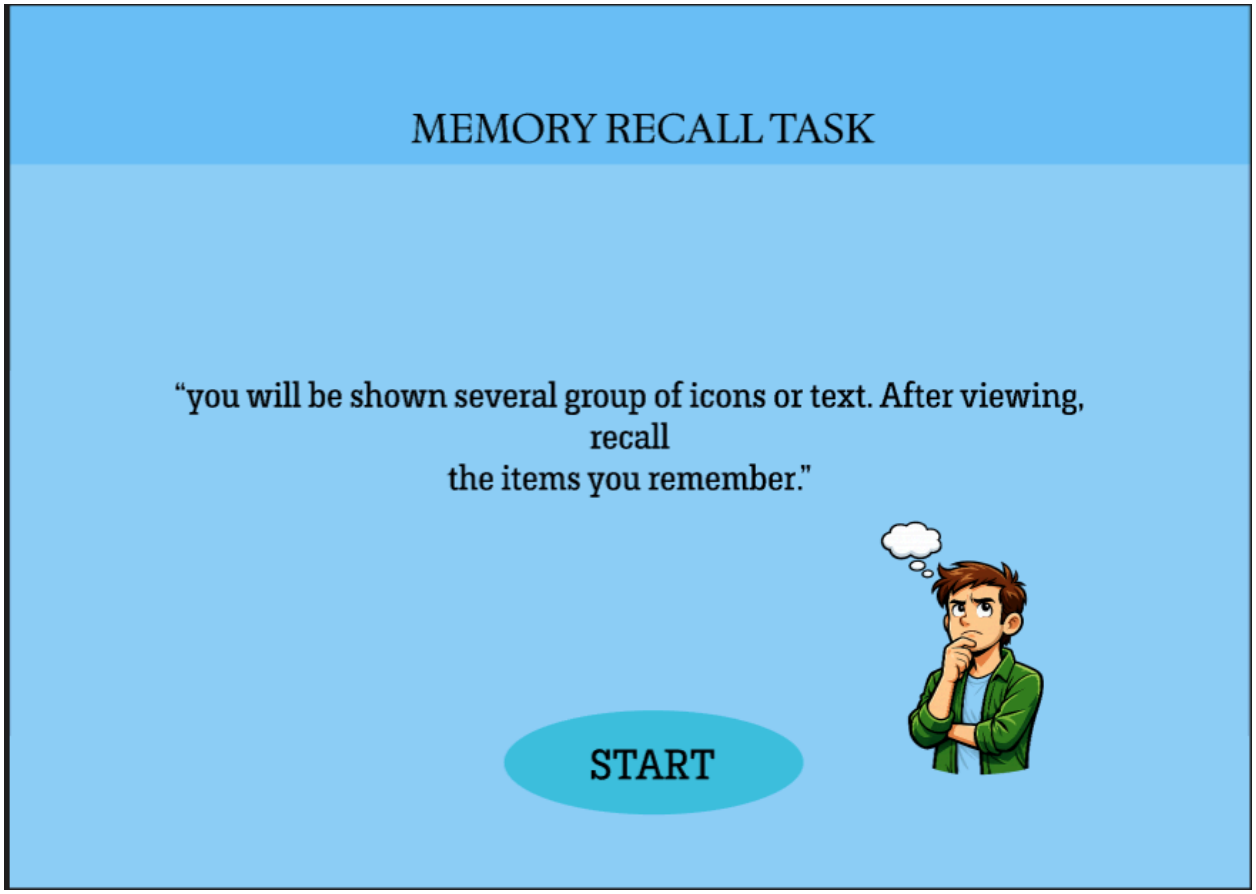
This experiment evaluates how chunked visual elements affect user recall performance.

DESIGN DESCRIPTION:

A. Home Screen

The home screen contains:
• Title: "Memory Recall Task"
• Instructions explaining the task
• Viewing duration (5 seconds)
• Start button to begin the experiment

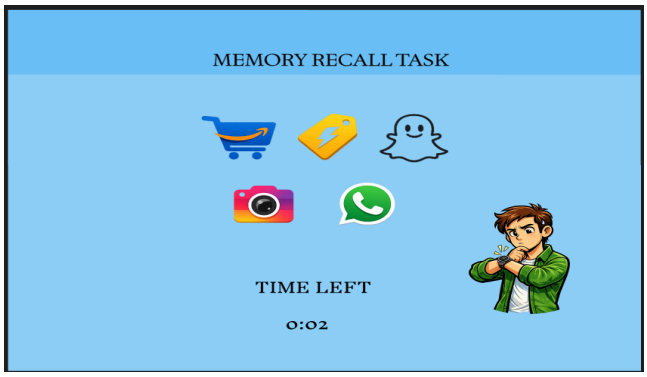
Purpose: To provide clear instructions before beginning the memory task.



B. Chunking Phase

The chunking phase displays:
- Five visual icons (Amazon, Flipkart, Snapchat, Instagram, WhatsApp)
- Items grouped visually
- A countdown timer from 5 seconds

Purpose: To allow users to observe and encode visual information into short-term memory.

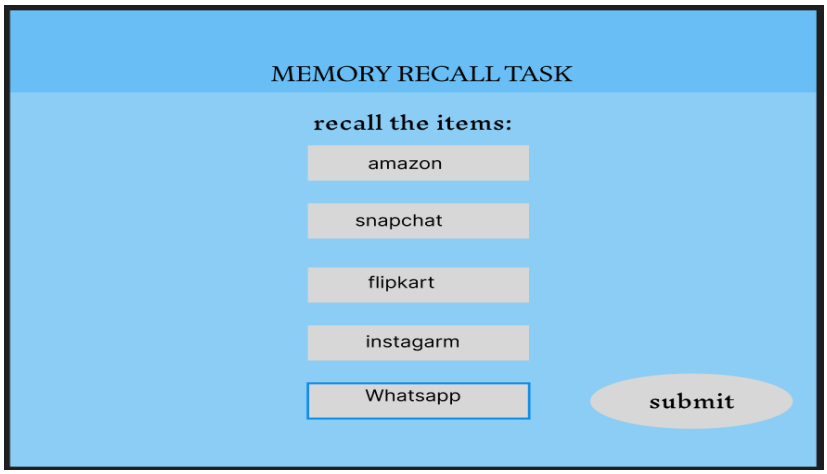


C. Recall Phase

The recall screen includes:

- Text prompting users to recall items
- Input fields to enter remembered items
- Submit button

Purpose: To measure recall accuracy after the viewing phase.



D. Result Screen

The result screen displays:
- Feedback message
- Score (e.g., 5/5 items correctly recalled)
- Home button

Purpose: To provide performance feedback to the user.

PROCEDURE:

- Created a Home Screen with instructions and Start button.

- Designed the Chunking Phase displaying grouped icons.
- Set a prototype delay of 5 seconds to simulate viewing time.
- Created a Recall Phase with input fields for user responses.
- Added a Submit button linked to the Result Screen.
- Displayed final score based on recall accuracy.

OBSERVATION:

Users were able to recall more items when the icons were visually grouped. Chunking reduced confusion and improved recognition patterns.

RESULT:

The experiment demonstrates that chunking significantly improves short-term memory recall performance.

CONCLUSION:

Chunking enhances memory retention by organizing information into structured groups. The experiment confirms that grouped visual elements improve recall accuracy compared to unstructured presentation.